SHYAMAPRASAD M. S.



DATA ENGINEER | PYTHON | PYSPARK | GOOGLE CLOUD | APACHE AIRFLOW

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PROFESSIONAL SUMMARY

Data Engineer / Python Developer with professional experience at Tata Consultancy Services, supporting data engineering work for an American Express client GenAI project. Hands-on experience with Python, SQL, PySpark, Google Cloud Platform, BigQuery, Dataproc, Cloud Storage, and Apache Airflow. Experienced in developing and maintaining data pipelines, building and consolidating Airflow DAGs, writing unit tests with Pytest and mocks, improving code coverage, and supporting data quality and production workflows. Strong academic record with an MCA (9.44 CGPA; 5th Rank, Kerala Technical University) and BCA (4th Rank, Kerala University).

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | **Software Engineer / Data Engineering — Bengaluru, India** **June 2025 – Present**

- Working on a GenAI project for the American Express client, contributing to data engineering and pipeline-related development and support.
- Developed and maintained Python/PySpark data-processing components and workflows involving data ingestion, transformation, and validation.
- Created unit tests using Pytest, mocks, and patching; improved test coverage of cs2dr_data_pull.py to 98.5%.
- Added and maintained tests for CECL and IFRS9 variance modules and raised pull requests for code changes.
- Consolidated US and international CCAR availability check DAGs into a single configurable DAG to reduce duplicated workflow logic.
- Worked with Google Cloud services and BigQuery, including SQL queries and updates to data/configuration tables; supported DAG monitoring and debugging.

TECHNICAL SKILLS

Programming:	Python, SQL, Java, C, C++, PySpark
Cloud & Data:	Google Cloud Platform (GCP), BigQuery, Google Cloud Storage (GCS), Dataproc, Pub/Sub
Data Engineering:	Data ingestion, ETL/data pipelines, data transformation, data validation, SQL development
Orchestration:	Apache Airflow, DAG development, scheduling, task dependencies, workflow monitoring
Testing & Quality:	Pytest, unit testing, mocking, patching, test coverage, debugging, code refactoring
Web & Database:	HTML, CSS, MySQL

SELECTED PROJECTS

- **Auto Claims Evaluator – Damage Score Analysis using ML and AI:** Built an academic web application using Python for the backend and HTML/CSS for the frontend, applying machine learning and AI concepts to support vehicle-damage assessment in auto-claims evaluation.
- **Online Payment Fraud Detection Using Machine Learning:** Developed an academic project using Python and HTML/CSS to explore machine-learning-based identification of potentially fraudulent online payment transactions.
- **Agroculture:** Created a web application intended to help agricultural officers and users access information about crops, fertilizers, irrigation, and pesticides.
- **Health Office Management System:** Developed a web-based application to automate selected services and workflows in a health office.

EDUCATION

2022 – 2024

Master of Computer Applications (MCA) — Lourdes Matha College of Science and Technology | **CGPA: 9.44/10** | **5th Rank, Kerala Technical University**

2019 – 2022

Bachelor of Computer Applications (BCA) — Christ Nagar College | **Score: 84.6%** | **4th Rank, Kerala University**

CERTIFICATIONS & RESEARCH

- NPTEL SWAYAM – Cloud Computing (65%) | NPTEL SWAYAM – Internet of Things (85%)
- Workshop Participation – Python Programming
- International Conference / Journal Certificates: Artificial Intelligence in Agriculture; Artificial Intelligence in the Medical Field; Disaster Management System using IoT